Josselin Davion-Jouffre

[email] | [phone] | [LinkedIn] | [GitHub]

6-month gap-year internship in quantitative finance / AI, from May 2027

Education

Mines Paris – PSL 2025 – 2029
MSc-level engineering degree (Ingénieur Civil) *Paris, France*

- Admitted ranked 24th in the national Mines-Ponts entrance exam (PSI track)
- Applied mathematics: stochastic calculus, probability, optimisation
- Data: data science, signal processing
- Gap year May 2027 – Aug 2028 before final year

Additional Training in ML for Quantitative Finance
Auditing student at ETH Zürich & self-study, alongside internship *Zurich, Switzerland*

- Audited course: *Machine Learning in Finance and Insurance* (Prof. P. Cheridito), with its 4 projects incl. deep hedging
- Personal programme: classical ML and boosting, financial time series (GARCH), options (Hull), purged cross-validation and backtesting (López de Prado), sequential deep learning
- Code on GitHub: [link to a public version of the repo]

Lycée Lakanal – Classes Préparatoires 2023 – 2025
PSI – intensive two-year programme in maths, physics & engineering science* *Sceaux, France*

Experience

ETH Zürich – Lab of Prof. Dirk Mohr Sep 2026 – Feb 2027
Research Intern, Machine-Learning-Assisted Mechanics *Zurich, Switzerland*

- Finite element simulations of the plastic behaviour of several materials (Abaqus Standard, Swift-Voce hardening laws)
- Dynamic split Hopkinson bar experiments and processing of experimental data (Python)
- Neural-network prediction of mechanical response (PyTorch, TensorFlow)
- Built a lab-wide Python tool (Sparrow nesting solver) generating optimised specimen cutting plans on new and partly used metal sheets: 4 sheets instead of 6 on a test campaign, and templates reused across campaigns to avoid recalibration costs at cutting suppliers

Colibri May 2026 – present
Co-founder

- Platform helping students declare private tutoring through the French home-services tax credit
- Built end-to-end with a co-founder: full-stack development (React, Supabase, French social-security (URSSAF) API integration), business-process design, sales, marketing
- 400 users, about €20k of tutoring processed per month, €1.3k monthly revenue

Projects

Mines Paris engineering project – CEMEF Nov 2025
Team project *Sophia Antipolis, France*

- Designed a hybrid electrode glass furnace with adaptive-gain PID temperature control: 30 % simulated energy saving over a conventional furnace geometry

Prépa research project (TIPE) – Phase-change material walls 2024 – 2025
Graded 20/20 (ENS entrance exam)

- Built an instrumented scale model (Arduino, temperature sensors, custom cooling and ventilation loop) and a finite element model of melting fronts in a thermal-storage wall

[Quant / ML project] [20XX]
[Context: personal, course, hackathon]

- [What you did, method, quantified result, GitHub link]

Skills

Programming: Python (NumPy, scikit-learn, PyTorch, TensorFlow), SQL, Git, LaTeX
Languages: French (native), English (C1), German (A2)
Interests: Programming, electronics and DIY (Arduino)